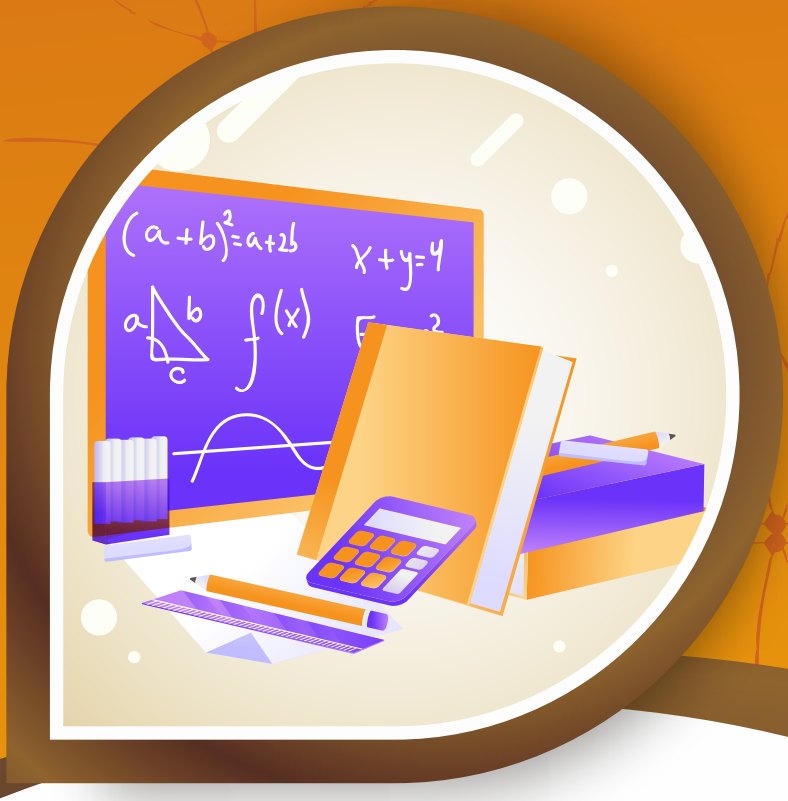
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

JEE Main Topic

(English Medium)

CHECK YOUR GRASP PRINCIPLE OF MATHEMATICAL INDUCTION EXERCISE-I

1. Let P(n) : $n^2 + n$; $n \in \mathbb{N}$ is an odd integer. It is seen that truth of P(n) $\Rightarrow$ the truth of P(n + 1). Therefore, P(n) is true for all-
(1) $n > 1$ (2) n
(3) $n > 2$ (4) None of these
MI0001

2. If $n \in \mathbb{N}$, then $x^{2n-1} + y^{2n-1}$ is divisible by-
(1) x + y (2) x – y (3) $x^2 + y^2$ (4) $x^2 + xy$
MI0002

3. If $n \in \mathbb{N}$, then $11^{n+2} + 12^{2n+1}$ is divisible by-
(1) 113 (2) 123
(3) 133 (4) None of these
MI0003

4. If $n \in \mathbb{N}$, then $3^{4n+2} + 5^{2n+1}$ is a multiple of-
(1) 14 (2) 16 (3) 18 (4) 20
MI0004

5. For every positive integer

n, $\frac{n^7}{7} + \frac{n^5}{5} + \frac{2n^3}{3} - \frac{n}{105}$ is-

(1) an integer
(2) a rational number which is not an integer
(3) a negative real number
(4) an odd integer
MI0005

6. Sum of the infinite seriese $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \ldots$ equals-
(1) 1/3 (2) 3 (3) 1/4 (4) 2
MI0006

7. If $a_k = \frac{1}{k(k+1)}$; then $\left(\sum_{k=1}^{n} a_k\right)^2$ is equal to-

(1) $\frac{n^2}{(n+1)^2}$ (2) $\frac{n^4}{(n+1)^4}$ (3) $\frac{n^2}{n^2+1}$ (4) $\frac{n}{n+1}$
MI0007

8. The sum of n terms of the series

$\frac{\frac{1}{2}\cdot\frac{2}{2}}{1^3} + \frac{\frac{2}{2}\cdot\frac{3}{2}}{1^3+2^3} + \frac{\frac{3}{2}\cdot\frac{4}{2}}{1^3+2^3+3^3} + \ldots\ldots$ is-

(1) $\frac{1}{n(n+1)}$ (2) $\frac{n}{n+1}$ (3) $\frac{n+1}{n}$ (4) $\frac{n+1}{n+2}$
MI0008

9. For all $n \in \mathbb{N}$, $7^{2n} - 48n - 1$ is divisible by-
(1) 25 (2) 26 (3) 1234 (4) 2304
MI0009

10. For all positive integral values of n, $3^{2n} - 2n + 1$ is divisible by-
(1) 2 (2) 4 (3) 8 (4) 12
MI0010

11. The smallest positive integer n for which the statement $3^{n+1} < 4^n$ holds is-
(1) 1 (2) 2 (3) 3 (4) 4
MI0011

12. For positive integer n, $10^{n-2} > 81n$ when-
(1) $n < 5$ (2) $n > 5$ (3) $n \geq 5$ (4) $n > 6$
MI0012

13. If P is a prime number then $n^p - n$; $n \in \mathbb{N}$ is divisible by p when n is a
(1) All natural number greater than or equal to 1
(2) odd number only
(3) even number only
(4) None of these
MI0013

14. A student was asked to prove a statement by induction. He proved
(i) P(5) is true and
(ii) Truth of P(n) $\Rightarrow$ truth of p(n + 1), $n \in \mathbb{N}$
On the basis of this, he could conclude that P(n) is true for
(1) no $n \in \mathbb{N}$ (2) all $n \in \mathbb{N}$
(3) all $n \geq 5$ (4) None of these
MI0014

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	4	1	3	1	1	1	1	2	4	1	4	3	1	3	

BRAIN TEASERS PRINCIPLE OF MATHEMATICAL INDUCTION EXERCISE-II

1. If $x > -1$, then the statement
$P(n) : (1 + x)^n > 1 + nx$ is true for-
(1) all $n \in \mathbb{N}$ (2) all $n > 1$; $n \in \mathbb{N}$
(3) all $n > 1$; $n \in \mathbb{N}$ and $x \neq 0$ (4) None of these
MI0015

2. For every positive integral value of n, 3^n, $> n^3$ when-
(1) $n > 2$ (2) $n \geq 3$ (3) $n \geq 4$ (4) $n < 4$
MI0016

3. $P(n) : 3^{2n+2} - 8n - 9$ is divisible by 64, is true for-
(1) all $n \in N \cup \{0\}$ (2) only for $n \geq 2$, $n \in N$
(3) $n \in N$, $n > 2$ (4) None of these
MI0017

4. If only for m, n are any two odd positive integer wih $n < m$, then the largest positive integers which divides all the numbers of the type $m^2 - n^2$ is-
(1) 4 (2) 6 (3) 8 (4) 9
MI0018

5. For all $n \in N$, $\cos\theta \cos2\theta \cos4\theta \ldots \cos2^{n-1}\theta$ equals to

(1) $\frac{\sin 2^n\theta}{2^n \sin\theta}$ (2) $\frac{\sin 2^n\theta}{\sin\theta}$

(3) $\frac{\cos 2^n\theta}{2^n \cos 2\theta}$ (4) $\frac{\cos 2^n\theta}{2^n \sin\theta}$ **MI0019**

6. $x(x^{n-1} - na^{n-1}) + a^n(n - 1)$ is divisible by $(x - a)^2$ for-
(1) $n > 1$ (2) $n > 2$
(3) all $n \in N$ (4) None of these
MI0020

7. For any odd integer $n \geq 1$
$n^3 - (n - 1)^3 + \ldots. + (-1)^{n-1}.1^3$ is equal to-

(1) $\frac{1}{4}(n + 1)^2(2n - 1)$ (2) $\frac{1}{4}(n - 1)^2(2n - 1)$

(3) $\frac{1}{2}(n - 1)^2(2n - 1)$ (4) $\frac{1}{2}(n + 1)^2(2n - 1)$
MI0021

8. If p and q are respectively. The sum and the sum of squares of n successive integers beginning with a, then $nq - p^2$ is-
(1) independent of a (2) independent of n
(3) dependent on a (4) None of these
MI0022

9. The sum of first n terms of the given series
$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots.$ is $\frac{n(n+1)^2}{2}$, when n is even. When n is odd, then sum will be- **[AIEEE-2004]**

(1) $\frac{n(n+1)^2}{2}$ (2) $\frac{1}{2}n^2(n + 1)$
(3) $n(n + 1)^2$ (4) None
MI0023

10. Let $S(k) = 1 + 3 + 5 + \ldots\ldots + (2k - 1) = 3 + k^2$, then which of the following is true ? **[AIEEE-2004]**
(1) S(1) is true
(2) $S(k) \Rightarrow S(k + 1)$
(3) $S(k) \not\Rightarrow S(k + 1)$
(4) Principle of mathematical Induction can be used to prove that formula **MI0024**

11. The sum of n terms of the series
$1 + (1 + a) + (1 + a + a^2) + (1 + a + a^2 + a^3) + \ldots..$, is-

(1) $\frac{n}{1-a} - \frac{a(1-a^n)}{(1-a)^2}$ (2) $\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$

(3) $\frac{n}{1-a} + \frac{a(1+a^n)}{(1-a)^2}$ (4) $-\frac{n}{1-a} + \frac{a(1-a^n)}{(1-a)^2}$
MI0025

12. **Statement -1 :** For every natural number $n \geq 2$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \ldots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

Statement –2 : For every natural number $n \geq 2$, $\sqrt{n(n+1)} < n+1$. **[AIEEE-2008]**
(1) Statement –1 is false, Statement –2 is true
(2) Statement–1 is true, Statement–2 is false
(3) Statement–1 is true, Statement–2 is true; Statement–2 is a correct explanation for Statement–1
(4) Statement–1 is true, Statement–2 is true; Statement–2 is not a correct explanation for Statement–1 **MI0026**

13. **Statement - 1:** For each natural number n, $(n + 1)^7 - n^7 - 1$ is divisible by 7.
Statement - 2: For each natural number n, $n^7 - n$ is divisible by 7. **[AIEEE-2011]**
(1) Statement-1 is false, statement-2 is true.
(2) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.
(3) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.
(4) Statement-1 is true, statement-2 is false.
MI0027

14. Consider the statement : "P(n): $n^2 - n + 41$ is prime." Then which one of the following is true? **[JEE(Main) Jan-19]**
(1) P(5) is false but P(3) is true
(2) Both P(3) and P(5) are false
(3) P(3) is false but P(5) is true
(4) Both P(3) and P(5) are true **MI0028**

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Ans.	3	3	1	3	1	3	1	1	2	2	1	3	2	4

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Important Notes

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